TraceSmith — GPU Profiling & Replay Architecture

Applications, pybind11 binding, C++17 core, GPU backends, and output formats

Dashed frame = architecture zone
Solid frame = subsystem
Box = module and responsibility

PROJECT OVERVIEW

TraceSmith is a cross-platform GPU profiling and deterministic replay system for AI compiler, framework, and driver engineers.
Its architecture separates applications, the pybind11 binding, a C++17 core, GPU backends, and output formats.

APPLICATION · user entry

- **C++ Application** — in-process capture
- **Python Application** — in-process capture
- **C++ CLI** — tracesmith; profile · record · view
- **Python CLI** — tracesmith-cli; export · analyze · replay
- **External UI** — Perfetto UI; Chrome tracing

BINDINGS · pybind11 boundary

- **_tracesmith.so** — python/src/bindings.cpp; pybind11 native extension

CORE LIBRARY · C++17

Cluster Module

- **GPUTopology** — NVLink · PCIe · NVML
- **MultiGPUProfiler** — cross-device capture
- **TimeSync** — NTP / PTP / CUDA
- **NCCLTracker** — collective ops
- **ClockCorrelator** — clock alignment
- **CommAnalysis** — communication matrix

Capture Module

- **CUPTIProfiler** — NVIDIA CUDA
- **MCPTIProfiler** — MetaX MACA
- **MetalProfiler** — Apple Metal
- **MemoryProfiler** — GPU memory
- **AscendProfiler** — Huawei CANN
- **createProfiler()** — nsys · mcTracer; xctrace · msprof

State Module

- **GPUStateMachine** — state analysis
- **InstructionStream** — dependency graph
- **TimelineBuilder** — ordered events
- **TimelineViewer** — ASCII rendering
- **PerfettoExporter** — JSON
- **PerfettoProtoExporter** — Protobuf

Replay Module

- **ReplayEngine** — core orchestrator
- **StreamScheduler** — operation order
- **OperationExecutor** — launch replay
- **DeterminismChecker** — result compare
- **FrameCapture** — RenderDoc-style

Common Module

- **Types** — TraceEvent · EventType; KernelParams
- **RingBuffer** — lock-free SPSC
- **StackCapture** — libunwind
- **XRayImporter** — LLVM XRay
- **Timestamps** — ns precision
- **TraceMetadata** — DeviceInfo

Format Module

- **SBTWriter** — binary writer
- **SBTReader** — binary reader
- **.sbt** — header + events; counters

Configuration

- **ProfilerConfig** — buffer · overflow
- **ReplayConfig** — mode · speed
- **StackCaptureConfig** — stack policy
- **NCCLTrackerConfig** — collective policy

GDB Module

- **RSPHandler** — GDB protocol
- **ProcessController** — ptrace / Mach
- **GPUDebugEngine** — breakpoints · replay
- **tracesmith-gdbserver** — v0.10.0

GPU BACKENDS · system interfaces

- **NVIDIA CUDA** — CUPTI SDK
- **Apple Metal** — Metal API
- **MetaX MACA** — MCPTI SDK
- **NCCL** — distributed
- **Nsight Systems** — nsys
- **mcTracer** — MetaX
- **xctrace** — Apple
- **libunwind** — host stacks
- **Perfetto SDK** — trace export

OUTPUTS · trace artifacts

- **trace.sbt** — binary format
- **trace.json** — Perfetto JSON
- **trace.perfetto** — Protobuf
- **graph.dot** — Graphviz
- **ASCII Timeline** — terminal
- **Comm Matrix** — heatmap JSON

The C++17 core contains capture, state, format, replay, cluster, configuration, and GDB. Applications, bindings, backends, and outputs are its boundaries.